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Does Structured Triage Context Help a Chest-Radiograph Pneumonia Model? A Time-Safe, Multi-Seed, Externally-Validated Multimodal Study with Missing-Data Cautions

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Abstract Background: Emergency-department (ED) triage records structured vital signs before chest imaging, motivating multimodal models that fuse the chest radiograph (CXR) with clinical context for pneumonia detection. Whether such fusion helps, under strict time-safety and rigorous multi-seed evaluation, remains unclear, and prior studies often report single-run results without leakage or missing-data audits. **Methods:** We construct a temporally safe cohort by linking MIMIC-CXR-JPG, MIMIC-IV and MIMIC-IV-ED, anchoring every feature at the CXR acquisition time t_0 and admitting only measurements resolved at or before t_0. On a patient-level temporal split (test n=1,075; prevalence 45.3%) we train image-only, concatenation- and attention-fusion, and clinical-baseline models across five random seeds, and add a time-safe early-laboratory panel as a third modality. We report AUROC, AUPRC, expected calibration error (ECE) and Brier score as mean±SD over seeds with paired per-seed deltas, patient-level bootstrap intervals, decision-curve and subgroup analyses, SHAP and Grad-CAM interpretability, external validation on NIH ChestX-ray14, and explicit temporal, split and preprocessing leakage audits. **Results:** A fine-tuned DenseNet-121 reaches AUROC 0.737 ± 0.003, far above triage-only baselines (0.61). Adding triage vitals leaves discrimination unchanged (ΔAUROC $+0.004 \pm 0.009$, within a pre-specified ± 0.05 margin). A calibration advantage reported from a single checkpoint (ECE 0.040 vs 0.067) shrinks to ΔECE -0.013 ± 0.016 across seeds; single-seed evaluation overstated it $\sim 2\times$. Adding early labs yields a small, seed-consistent AUROC gain ($+0.009 \pm 0.005$) that an ablation shows is reproduced exactly by lab *missingness indicators alone*, with no contribution from lab values, and is not leakage. Externally, discrimination transfers with minimal loss (AUROC 0.722 ± 0.005 on 112,120 NIH radiographs). **Conclusion:** The chest radiograph carries the discriminative signal; immediately available clinical context does not reliably add discrimination, and naive multimodal fusion can appear beneficial for reasons that will not generalise. We offer two transferable cautions for clinical multimodal AI: single-seed calibration inflation, and missingness masquerading as multimodal benefit.

Keywords Pneumonia detection · Chest radiograph · Multimodal fusion · Emergency department triage · Calibration · Reproducibility · Missing data · External validation

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1 Introduction

Pneumonia is an acute lower respiratory tract infection and a leading cause of emergency-department (ED) presentation, hospital admission and death worldwide (Metlay et al, 2019). The frontal chest radiograph (CXR) is the recommended first-line imaging investigation, but its clinical value is bottlenecked by radiologist availability during peak ED demand. Automated CXR analysis that produces *well-calibrated* probability estimates is therefore an attractive decision-support tool, and deep convolutional networks now approach radiologist-level performance on several radiographic findings (Huang et al, 2017).

A structurally appealing but underused source of complementary information is the ED triage record. On arrival, a triage nurse records vital signs (temperature, heart rate, respiratory rate, peripheral oxygen saturation (SpO_2), blood pressure, a pain score) and an acuity level, *before* any imaging is ordered. Multimodal models that fuse the CXR with this immediately available context are widely proposed (Acosta et al, 2022; Hayat et al, 2022; Soenksen et al, 2022), on the intuition that combining the two streams should better reflect the clinical picture at the moment of decision.

Three methodological issues, however, recur across this literature and motivate the present study. First, *temporal safety*: clinical variables are frequently included without verifying they were actually available at prediction time, risking optimistic leakage from information generated after the decision point. Second, *single-seed reporting*: calibration and small discrimination effects are often reported from a single training run, although both are sensitive to training stochasticity (Guo et al, 2017; Nixon et al, 2019). Third, *missing data*: clinical inputs in the ED are heavily and non-randomly missing, so a model may exploit *whether* a measurement exists rather than its value, an effect that can masquerade as a genuine multimodal benefit.

We address a deliberately conditional question, *when, if ever, does immediately available clinical context help a CXR pneumonia model?*, under strict time-safety, with multi-seed replication and explicit missing-data and leakage audits, and report in line with prediction-model and medical-imaging guidance (Collins et al, 2024; Tejani et al, 2024). Our contribution is not a new architecture but a rigorous, reproducible, externally-validated evaluation that yields a clear answer and two transferable cautions:

- A **temporally safe, reproducible multimodal ED cohort** linking three MIMIC databases at the CXR acquisition time t_0, with every feature verified to precede imaging (0 of 314,145 laboratory values post-t_0).
- A **multi-seed evaluation** (five seeds) of image-only, triage-fusion and laboratory-fusion models with paired bootstrap comparisons, showing that immediately available clinical context is *discrimination-neutral.*
- **Caution 1 (single-seed calibration inflation):** a calibration advantage that appears large from one checkpoint shrinks roughly twofold and becomes seed-fragile under replication.
- **Caution 2 (missingness as signal):** a small "labs" gain that is reproduced entirely by laboratory *missingness indicators*, not the measured values, and is not attributable to leakage.
- **External validation** of the image model on 112,120 NIH ChestX-ray14 radiographs, showing cross-institution generalisation with an AUROC drop of only 0.016, together with subgroup, decision-curve, interpretability and modern-backbone analyses.

2 Background and Related Work

2.1 Pneumonia and ED triage

Pneumonia manifests on a frontal CXR as regions of increased opacity (consolidation) where air-filled alveoli fill with exudate. Reported CXR sensitivity for pneumonia is moderate (0.43–0.74) with high specificity, reflecting both the difficulty of early detection and reliable exclusion in unambiguous cases (Metlay et al, 2019). Triage vital signs quantify physiological perturbation and severity; they are highly correlated with pneumonia but non-specific, since tachycardia, hypoxia and high acuity accompany many respiratory presentations. Crucially, triage measurements are recorded before imaging is ordered (a workflow guarantee, not a statistical regularity) so they carry no risk of temporal leakage.

2.2 The MIMIC database family

The Medical Information Mart for Intensive Care (MIMIC) programme provides the largest publicly available de-identified clinical dataset linked to real hospital records from the Beth Israel Deaconess Medical Center (BIDMC). This study uses three interoperable components. **MIMIC-CXR-JPG** (Johnson et al, 2019) contains 227,835 chest-radiograph studies from 64,588 patients (2011–2019) with CheXpert labels for 14 findings. **MIMIC-IV** (Johnson et al, 2023b) provides hospital-admission and laboratory records, enabling temporal consistency checks and the laboratory modality. **MIMIC-IV-ED** (Johnson et al, 2023a) provides ED-specific records for all BIDMC emergency visits, including the structured triage table used as the tabular input. All are distributed through PhysioNet under a Data Use Agreement; per-patient date shifts preserve temporal ordering and durations.

2.3 Automated labelling: CheXpert

CheXpert (Irvin et al, 2019) is an NLP pipeline that extracts binary labels for 14 thoracic conditions from free-text reports, assigning each a value of positive, negative, uncertain (-1), or not-mentioned. The uncertain label captures hedging language ("cannot exclude", "possible pneumonia"). We adopt *u_ignore* (exclude uncertain studies) as the primary policy, which preserves label quality at the cost of a smaller, higher-prevalence evaluation set.

2.4 Deep learning and multimodal fusion

DenseNet-121 (Huang et al, 2017) propagates features through dense connectivity, $x_\ell = H_\ell([x_0, \ldots, x_{\ell-1}])$, and forms a strong CXR backbone under transfer learning. Intermediate (feature-level) fusion extracts modality-specific embeddings and combines them, by concatenation or by attention over modality tokens (Acosta et al, 2022).

Prior clinical fusion studies typically report positive gains, but on tasks that differ from ours in an important way. MedFuse fuses MIMIC clinical time-series with CXR for ICU mortality and phenotyping (Hayat et al, 2022), and the HAIM framework fuses imaging, tabular, text and time-series for inpatient outcomes (Soenksen et al, 2022); both target *outcomes* that are not directly visible in the radiograph. Recent emergency-department work is similar: Lee et al. (Lee et al, 2024) fuse CXR and EHR to screen for acute heart failure in the ED (AUROC 0.89), again a task whose label is not encoded in the image. Our result, that fusion is discrimination-neutral for radiographic *pneumonia*, is consistent with rather than contradictory to this body of work: pneumonia is defined *by* the radiograph, so the image already contains the label and there is little room for clinical context to add discriminative signal. This distinction, that fusion tends to help when the target lies partly outside the image but not when the image is essentially the target, is a useful lens on when multimodal CXR fusion should be expected to help.

Two further literatures frame our cautions. *Informative missingness*, the observation that missing values and their patterns correlate with the label, is well established and is often *exploited* as a predictive feature (Che et al, 2018); we demonstrate the flip side, that within a fusion model it can *masquerade* as a chemistry-driven benefit and therefore must be controlled for. *Calibration under distribution shift* is known to be fragile for single models and post-hoc scaling, with model averaging more robust (Ovadia et al, 2019); this is consistent both with our single-seed calibration caution and with the prevalence-driven degradation of ECE we observe on the external NIH set.

To our knowledge, no prior work simultaneously (i) targets ED-specific pneumonia, (ii) uses ED triage vitals under a verified time-safety constraint, (iii) evaluates with a patient-level temporal split *and* multi-seed replication, and (iv) audits missingness and leakage explicitly. This study fills that gap.

3 Materials and Methods

3.1 Datasets and how they are used

The task is binary radiographic-pneumonia classification of a frontal ED CXR. Each of the three MIMIC components plays a distinct role. MIMIC-CXR-JPG supplies the *images* (JPEG radiographs) and the *labels* (CheXpert pneumonia), together with DICOM metadata (view position; StudyDate/StudyTime, from which the temporal anchor t_0 is derived). MIMIC-IV-ED supplies the *primary tabular modality*: the triage table (vitals, pain, acuity) recorded at ED intake, linked to each CXR through the ED stay. MIMIC-IV supplies the *admission records* used to verify temporal consistency and the *laboratory events* used for the time-safe laboratory modality. The image branch consumes MIMIC-CXR-JPG; the tabular branch consumes MIMIC-IV-ED (triage) and, in the laboratory variant, MIMIC-IV (labs); the CheXpert labels supervise all models.

3.2 Cohort construction pipeline

A scripted pipeline links the sources through temporally constrained relational joins. Each CXR study is linked to a hospital admission (by `subject_id` and admission window) and to an ED stay (by `stay_id` with $t_0 \in [\text{intime}, \text{outtime}]$); one-to-one study-to-stay integrity is enforced by deduplication. The triage table is merged on `stay_id`; studies are restricted to frontal (PA/AP) projections; CheXpert pneumonia labels are merged and the *u_ignore* policy applied. The temporal anchor is $t_0 = \text{StudyDate} + \text{StudyTime}$ from DICOM metadata. The resulting cohort comprises 9,154 ED-anchored studies; the patient-level temporal split (below) assigns train 7,144 / validation 930 / test 1,080 studies. Models train and evaluate only on radiographs present on disk in our data snapshot, yielding a held-out test set of n=1,075 (487 positive / 588 negative; prevalence 45.3%). The entire pipeline is reproducible from a single entry point, emitting versioned manifests at each stage. Table 1 summarises the cohort: the split sizes, test-set demographics (median age 65; 55% female; predominantly White with substantial Black and Hispanic representation), the near-even PA/AP view split, and the triage vital-sign distributions with their missingness rates (3–8% per vital), which motivate the per-vital missingness indicators.

3.3 Time-safety and the patient-level temporal split

Every clinical feature is either a triage measurement (recorded at intake, before imaging by protocol) or a laboratory value with charttime in $(t_0 - 24\,\mathrm{h}, t_0]$. Time-safety was verified empirically on the extracted laboratory events: *zero* of 314,145 measurements have a charttime after t_0 (median lead time 1 h). Patients are ranked by their earliest t_0 and partitioned 80/10/10 into train/validation/test, so no patient appears in more than one split (verified: 0 cross-split subjects, 0 duplicate studies). This ordinal-time construction approximates prospective deployment. The tabular preprocessor (median imputation, standardisation, one-hot encoding) is fit on the training split only.

3.4 Feature engineering

Triage features are temperature, heart rate, respiratory rate, SpO_2, systolic and diastolic blood pressure, pain score and ESI acuity, plus the radiograph view and demographics. Physiologically implausible values are clipped to clinical ranges. For each vital a binary *missingness indicator* is added and the value median-imputed; because ED missingness is not missing-at-random, the indicator itself carries information. **Laboratory features** comprise 26 concepts (complete blood count, metabolic panel, liver panel, blood gas, lactate, and the inflammatory markers CRP and procalcitonin). For each study and concept we take the last value resolved at or before t_0 within the 24 h window, and add a missingness flag recomputed after the cohort join.

3.5 Models

The **image model** is DenseNet-121, initialised from ImageNet, multilabel-pretrained on 182,637 non-ED MIMIC-CXR studies (14 CheXpert findings, masked binary cross-entropy over observed labels), then fine-tuned for binary pneumonia at 224 × 224 with class-balanced loss and differential learning rates (5×10^{-5} head, 10^{-5} backbone). Multilabel pretraining minimises a masked binary cross-entropy that ignores not-mentioned CheXpert labels,

$$\mathcal{L}_{\mathrm{mBCE}} = -\frac{1}{\sum_i m_i} \sum_i m_i \left[y_i \log \hat{p}_i + (1-y_i) \log(1-\hat{p}_i) \right], \tag{1}$$

where $m_i \in \{0,1\}$ masks unobserved conditions. The triage vector is encoded by a two-layer TabularMLP (batch-normalised, dropout 0.2) to a 128-d embedding e_t. The **concatenation** model concatenates the 1024-d image embedding e_v with e_t, $[e_v; e_t] \in \mathbb{R}^{1152}$, and applies a fully connected head; the **attention** model projects each embedding to a d-dimensional token, prepends a learnable `CLS` token, and applies a two-layer Transformer whose scaled dot-product attention

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{2}$$

mixes the $[\mathtt{CLS}, e_v, e_t]$ tokens; the `CLS` output feeds the classifier (Acosta et al, 2022). The **+labs** model appends the 26 lab values and their flags to the tabular vector. **Clinical baselines** are logistic regression and XGBoost (Chen and Guestrin, 2016) on triage features. A **modern-backbone baseline** uses a CheXpert-trained DenseNet-121 (Cohen et al, 2022) (not trained on MIMIC), zero-shot and as a frozen-feature linear probe. All deep models share the multilabel-pretrained backbone so the multi-seed analysis isolates fine-tuning stochasticity.

3.6 Training and evaluation

Each deep model is trained across five seeds $\{42, 123, 456, 789, 1000\}$ with early stopping on validation AUPRC. Primary metrics are AUROC, AUPRC, ECE and Brier score, reported as mean±SD over seeds with paired per-seed deltas against the image-only model on identical test rows; within-checkpoint uncertainty uses a 2,000-replicate patient-level paired bootstrap (patients, not studies, are resampled). A non-inferiority margin $\Delta_0 = -0.05$ AUROC was pre-specified (Piaggio et al, 2012). Calibration is assessed with the ECE

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \left| \bar{p}_b - \bar{y}_b \right|, \tag{3}$$

Table 1 Cohort characteristics. Splits are patient-level and temporally ordered; demographic and vital-sign statistics are for the test split.

Characteristic	Value
Splits: studies / patients / prevalence	
Train	7,144 / 6,045 / 46.9%
Validation	930 / 787 / 48.8%
Test	1,080 / 890 / 45.4%
Test-set demographics	
Age, mean (median)	64 (65)
Female	55.0%
White / Black / Hispanic	678 / 258 / 71
Asian / Other	36 / 37
View: PA / AP	56.8% / 43.2%

Triage vital (test set)	mean±SD (% missing)
Temperature (°F)	98.5±1.4 (5.9%)
Heart rate (bpm)	87.7±19.6 (3.3%)
Respiratory rate (/min)	18.6±3.5 (4.5%)
SpO_2 (%)	97.4±3.6 (4.8%)
Systolic BP (mmHg)	135±24 (4.0%)
Diastolic BP (mmHg)	73±16 (4.3%)
Pain (0–10)	3.8±4.3 (8.2%)
Acuity (ESI 1–5)	2.3±0.7 (1.2%)

(with B=10 uniform bins; $\bar{p}_b, \bar{y}_b$ the mean predicted probability and observed frequency in bin b), the maximum calibration error, Brier score and reliability diagrams; ECE bin-count/scheme sensitivity is examined. Clinical utility uses decision-curve analysis with net benefit

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}}{n} - \frac{\mathrm{FP}}{n} \cdot \frac{p_t}{1-p_t}. \tag{4}$$

Interpretability uses exact TreeSHAP (Chen and Guestrin, 2016) for the XGBoost baseline and Grad-CAM for the image branch. We additionally compute subgroup metrics (sex, age, race, acuity, view), operating points at fixed sensitivity, a laboratory *missingness-only* ablation, and external validation on all 112,120 NIH ChestX-ray14 radiographs (Wang et al, 2017) with preprocessing matched to the internal transform.

4 Results

4.1 Discrimination: imaging dominates, fusion is neutral

Table 2 and Figs. 1–3 summarise the primary results. Both CXR models exceed the triage-only baselines by more than 12 AUROC points, confirming that the radiograph carries the discriminative signal. Adding triage vitals does not change discrimination: the paired concat−image ΔAUROC is $+0.004 \pm 0.009$ (range $[-0.007, +0.014]$), with the effect's lower bound well inside the pre-specified -0.05 non-inferiority margin. Attention fusion matches image AUROC but is severely and variably miscalibrated (ECE 0.076 ± 0.042; Fig. 4), illustrating that AUROC alone is insufficient for model selection. The precision-recall curves (Fig. 2) mirror the AUROC ordering at the 45.3% prevalence.

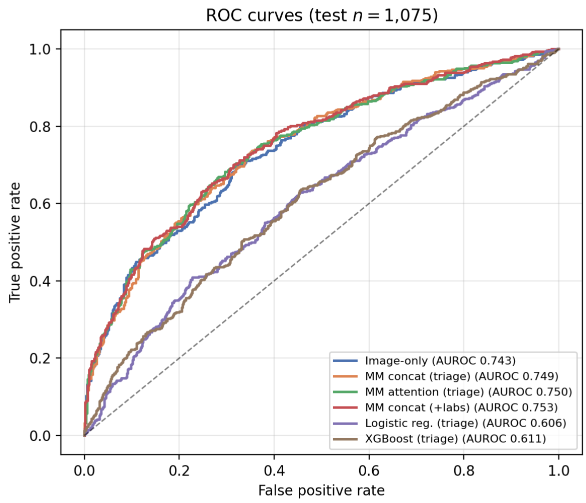

Fig. 1 ROC curves on the held-out test set (5-seed mean predictions for the deep models; AUROC of the ensemble curve shown, slightly above the per-seed means in Table 2).

4.2 Feature ablation

Retraining the concat model with restricted triage feature subsets (Table 3, representative seed 42) leaves AUROC essentially unchanged (0.736–0.752, a 0.016 range), confirming that the CXR backbone, not any particular triage feature, drives discrimination. Removing the missingness indicators does not reduce performance, consistent with their marginal, largely missingness-pattern contribution (Section 4.4).

4.3 Caution 1: single-seed evaluation inflates the calibration effect

A single checkpoint suggested the concat model was markedly better calibrated (ECE 0.040 vs 0.067, a 40% reduction). Across seeds, however, the image-only ECE alone ranges 0.041–0.060, and the paired advantage shrinks to ΔECE -0.013 ± 0.016 (4/5 seeds

Table 2 Multi-seed test-set performance (mean±SD over 5 seeds; n=1,075; prevalence 45.3%). Clinical baselines are single-fit.

Model	AUROC	AUPRC	ECE	Brier
Image-only DenseNet-121	0.737 ± 0.003	0.719 ± 0.004	0.053 ± 0.008	0.207 ± 0.001
Multimodal (concat, triage)	0.741 ± 0.006	0.715 ± 0.005	0.040 ± 0.014	0.205 ± 0.002
Multimodal (attention, triage)	0.738 ± 0.006	0.713 ± 0.012	0.076 ± 0.042	0.212 ± 0.010
Multimodal (concat, +labs)	0.747 ± 0.004	0.724 ± 0.004	0.043 ± 0.011	0.203 ± 0.002
Logistic regression (triage)	0.606	0.548	0.037	0.242
XGBoost (triage)	0.611	0.567	0.046	0.241

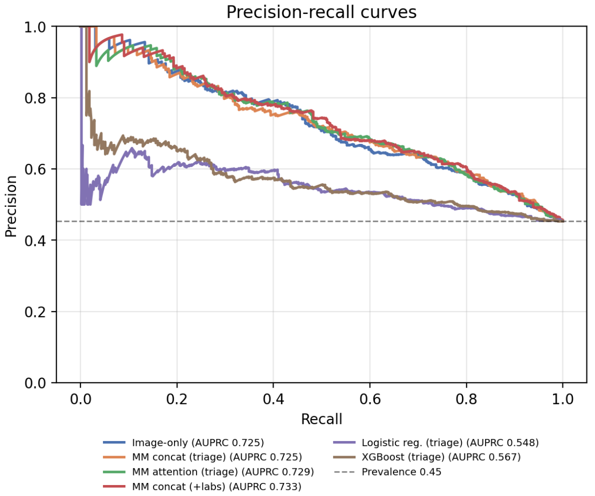

Fig. 2 Precision-recall curves at test prevalence 45.3%.

Table 3 Triage feature-group ablation for the concat model (seed 42). AUROC is stable across subsets, confirming the CXR backbone dominates.

Triage features	AUROC	AUPRC
All (vitals, acuity, demographics, flags)	0.736	0.714
Vitals only	0.740	0.714
Vitals + acuity	0.744	0.723
No missingness flags	0.752	0.734

favour concat; the range crosses zero). The single-seed estimate thus overstated the benefit roughly twofold. The apparent gap is also binning-dependent (Fig. 6): under coarser or quantile bins it shrinks further (Nixon et al, 2019). Brier scores are essentially unchanged, so any effect is distributional alignment rather than refinement. Reliability diagrams (Fig. 5) show all deep models tracking the diagonal closely, with attention fusion the clear outlier.

4.4 Caution 2: the "labs" benefit is missing-data structure

Appending early labs gives the only seed-consistent AUROC gain (labs−image +0.009±0.005, 5/5 seeds, range excluding zero). Three analyses show this is neither chemistry nor leakage. *Ablation:* a model using laboratory *missingness flags only* (no values) reproduces the gain exactly (flags−image +0.009; Fig. 4); the lab values add nothing further to discrimination. *Subcohort:* on the 269 test studies that actually have labs, discrimination is tied (labs 0.778 vs image 0.776). *Leakage audit:* 0 of 314,145 lab measurements post-t_0; 0 cross-split subjects; preprocessing fit on train only; the "any lab drawn" indicator has near-null univariate AUROC (0.464). The gain therefore reflects the weak complementary signal of *whether* bloodwork was ordered, is small, and lies within the patient-level bootstrap CI.

4.5 Clinical availability of laboratory results

Under time-safe matching, only 25.4% of ED-anchored CXR studies (272/1,075 test) have any resolved laboratory value at or before imaging (Fig. 7). Complete-blood-count and metabolic panels reach ∼25%, but the most pneumonia-specific markers (CRP and procalcitonin) are present in only 0.5% and 0.6% of studies. At the moment of imaging the informative labs are essentially unavailable, which explains both the small labs effect and its missingness origin.

4.6 Operating points and clinical utility

ED triage for pneumonia prioritises sensitivity. At a fixed sensitivity of 0.90 the specificity is 0.31 (image) versus 0.32 (concat); at 0.95, 0.17 versus 0.18 (Table 4), clinically equivalent operating behaviour. Decision-curve analysis (Fig. 8) shows both CXR models providing substantially higher net benefit than the triage-only baselines and the treat-all/treat-none references across the clinically relevant threshold range, with the image and multimodal curves essentially overlapping.

4.7 Subgroup performance

Table 5 and Fig. 9 report subgroup AUROC. Performance is markedly lower for female than male patients

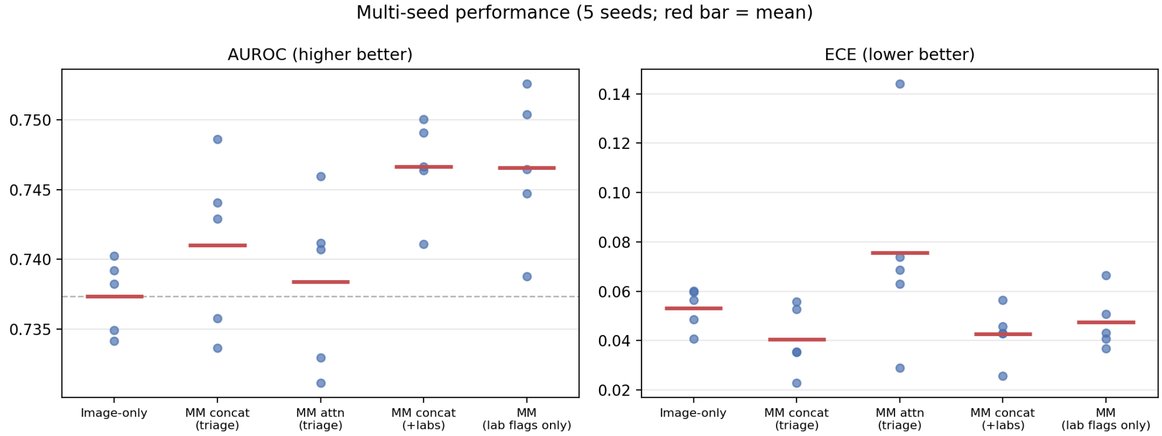

Fig. 3 Per-model AUROC and ECE across five seeds (points, seeds; red bar, mean). Discrimination is stable; the image-only ECE alone varies substantially by seed.

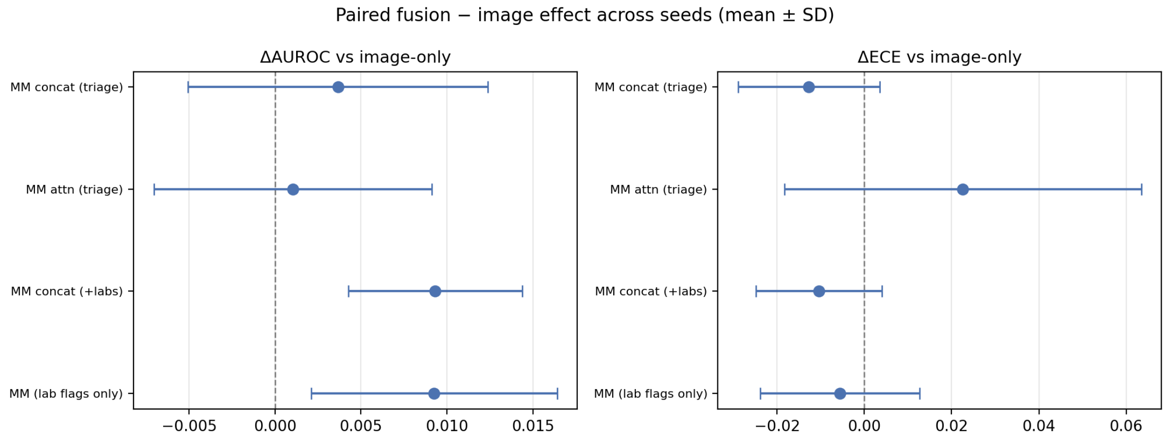

Fig. 4 Paired fusion−image effects across seeds (mean±SD). Every effect straddles zero; the labs and lab-flags-only deltas are identical.

Table 4 Operating points at fixed sensitivity (mean over 5 seeds).

Model	Sens.	Spec.	PPV	NPV
Image	0.90	0.310	0.520	0.791
Concat	0.90	0.322	0.524	0.797
Image	0.95	0.173	0.488	0.808
Concat	0.95	0.179	0.490	0.814

(0.71 vs 0.77) and for Hispanic patients (0.64) relative to White (0.75); PA views outperform AP (0.75 vs 0.72). Fusion does not consistently close these gaps and, in the smallest strata, can slightly worsen calibration. These disparities echo documented underdiagnosis biases in CXR models (Seyyed-Kalantari et al, 2021; Larrazabal et al, 2020) and are reported here as a prerequisite fairness audit rather than a solved problem; small minority strata carry wide confidence intervals.

Table 5 Subgroup AUROC (mean over 5 seeds) for image-only and concat fusion. Small strata (e.g. Hispanic, Asian) have wide CIs.

Subgroup	Image	Concat
Female	0.713	0.715
Male	0.770	0.776
White	0.751	0.756
Black	0.710	0.709
Hispanic	0.640	0.655
PA view	0.748	0.758
AP view	0.721	0.715
High acuity (1–2)	0.759	0.761
Lower acuity (3–5)	0.705	0.711

4.8 Interpretability

For the triage-only XGBoost model, exact TreeSHAP attributes the largest importance to SpO_2 (mean $|SHAP| = 0.148$), followed by heart rate (0.099), temperature (0.078) and respiratory rate (0.062) (Table 6;

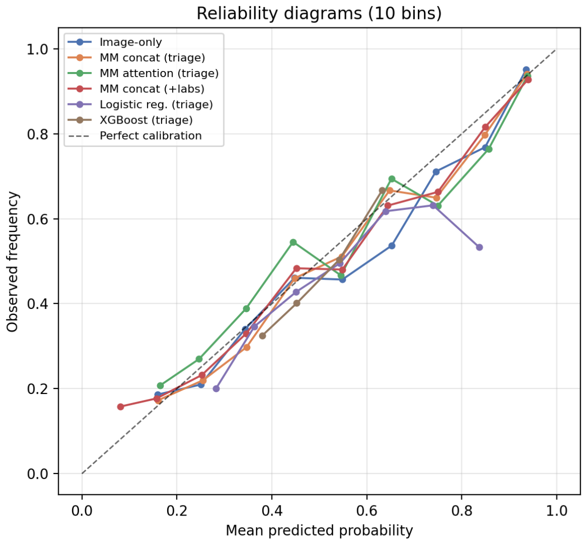

Fig. 5 Reliability diagrams (10 bins). The deep models track the diagonal; attention fusion systematically under-predicts.

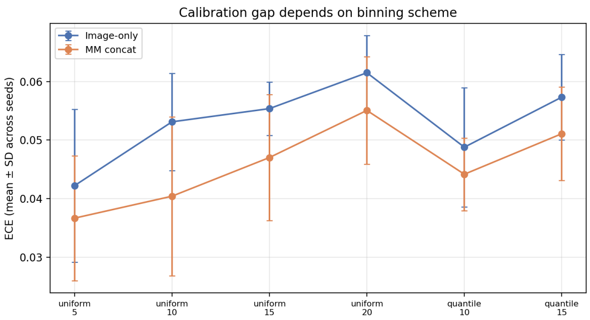

Fig. 6 ECE under different binning schemes; the image–concat calibration gap depends on the binning choice.

Table 6 Mean |SHAP| for the triage XGBoost model (test set).

Feature	Mean |SHAP|
SpO_2	0.148
Heart rate	0.099
Temperature	0.078
Respiratory rate	0.062
PA-view indicator	0.053
Systolic BP	0.043
Diastolic BP	0.042
Acuity (ESI)	0.031
Pain score	0.012

Fig. 10), consistent with the pathophysiology of pneumonia, in which alveolar consolidation impairs gas exchange. Grad-CAM on the image branch (Fig. 11) localises activation over regions of consolidation in confident true positives, whereas a false positive attends to overlapping opacity and a false negative shows diffuse, non-localised activation.

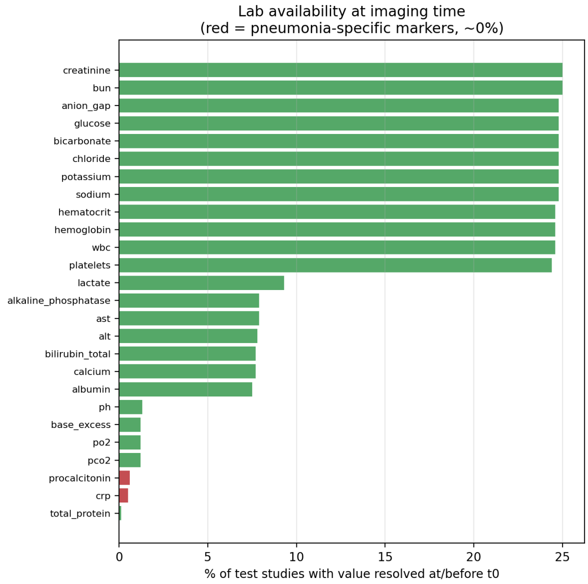

Fig. 7 Fraction of test studies with each laboratory value resolved at or before t_0; pneumonia-specific markers (red) are essentially unavailable at imaging time.

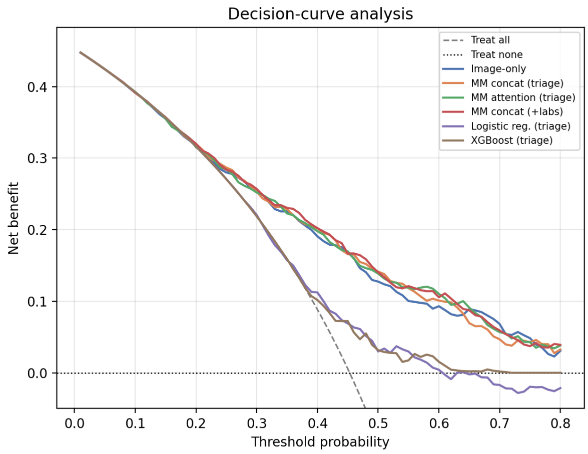

Fig. 8 Decision-curve analysis: net benefit versus threshold probability. Imaging models dominate the triage baselines and reference strategies.

4.9 Modern backbone and external validation

A CheXpert-trained DenseNet-121 (Cohen et al, 2022), used zero-shot (AUROC 0.667) or as a frozen-feature linear probe (0.687), does not outperform our task-tuned model (0.737), indicating the conclusions are not an artefact of an outdated backbone. On external validation against all 112,120 NIH ChestX-ray14 radiographs (Wang et al, 2017) (a different institution, scanner population and label pipeline; prevalence 1.28%), discrimination transfers with minimal loss: AUROC 0.722 ± 0.005 across five seeds (ensemble 0.726), only 0.016 below the internal 0.737 (Fig. 12). As ex-

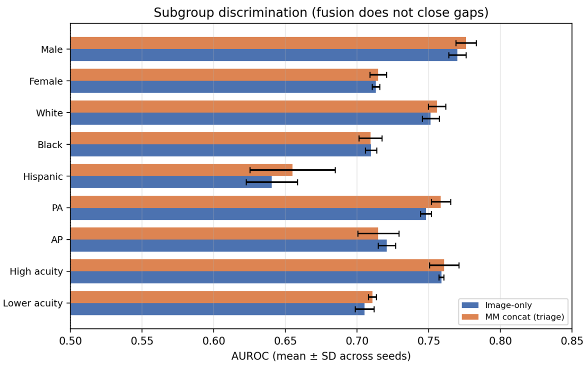

Fig. 9 Subgroup AUROC (mean±SD across seeds); fusion does not close sex, race or view gaps.

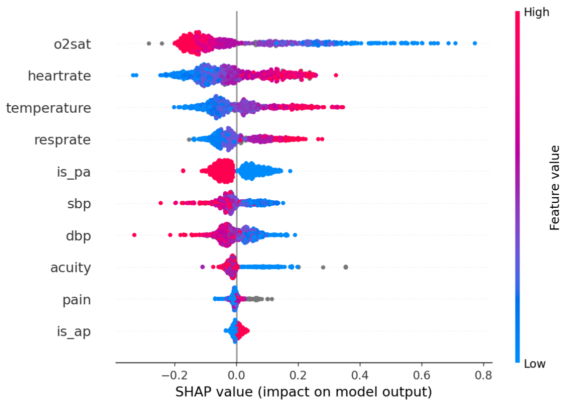

Fig. 10 SHAP summary (beeswarm) for the triage XGBoost model; low SpO_2 pushes predictions toward pneumonia.

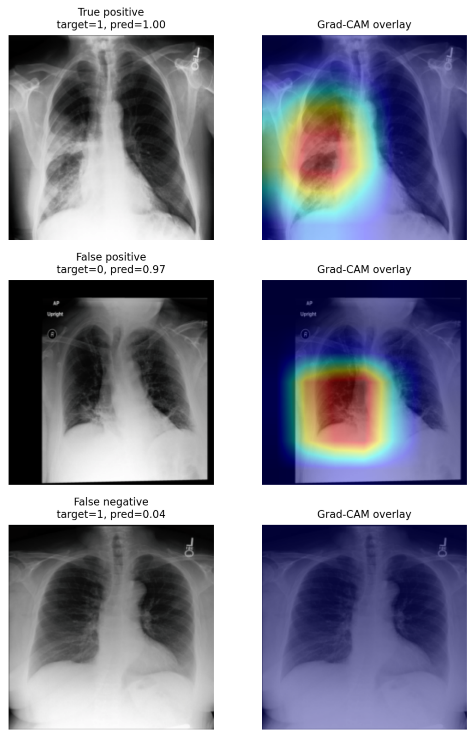

Fig. 11 Grad-CAM on the image model for representative test cases: a confident true positive, a false positive, and a false negative.

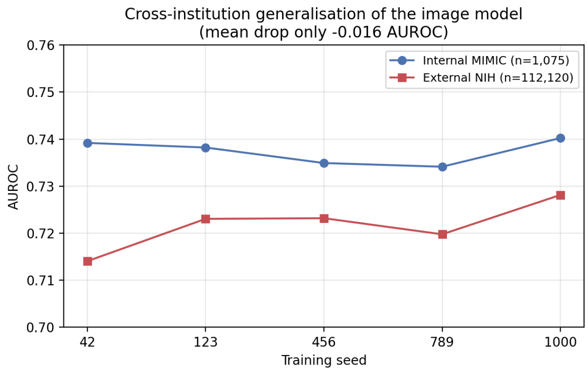

Fig. 12 Cross-institution generalisation: per-seed AUROC on internal MIMIC (n=1,075) versus external NIH ChestX-ray14 (n=112,120); mean drop −0.016.

pected under the large prevalence shift (45% → 1.3%), prevalence-dependent metrics degrade (AUPRC 0.030, ECE 0.44) because probabilities are calibrated to the internal base rate; AUROC, which is prevalence-independent, is the appropriate cross-dataset metric, and it holds despite domain shift and NLP-derived external labels.

4.10 Label-policy sensitivity

CheXpert uncertain labels can be excluded (*u_ignore*, our primary policy), mapped to negative (*u_zero*) or positive (*u_one*); each changes the cohort size and prevalence (Table 7). The *u_zero* policy sharply lowers AUROC and AUPRC because treating uncertain studies as negative injects label noise and false negatives, whereas *u_one* inflates prevalence to 69.8% and thus raises both metrics; *u_ignore* is intermediate and the most defensible. Critically, the image and concat models remain essentially equivalent under every policy, confirming that the discrimination-neutrality of fusion is not an artefact of the uncertainty-handling choice.

5 Discussion

The chest radiograph carries the discriminative signal for radiographic pneumonia in this ED cohort. Immediately available structured clinical context (triage vitals, and even a broad early-laboratory panel) does not add reliable discrimination, and the image model generalises externally with minimal loss. The asymmetry is intuitive: the radiograph encodes the outcome (radiographic consolidation) almost directly, whereas vital signs and labs encode severity and physiological per-

Table 7 Label-policy sensitivity (seed 42): AUROC/AUPRC for image and concat under three CheXpert uncertainty policies. Test size and prevalence differ by policy.

Model	Policy	Test n	AUROC	AUPRC
Image	u_ignore	1,075	0.739	0.724
	u_zero	1,944	0.597	0.342
	u_one	1,944	0.761	0.873
Concat	u_ignore	1,075	0.736	0.714
	u_zero	1,944	0.595	0.357
	u_one	1,944	0.765	0.876

turbation that are correlated with, but not specific to, pneumonia.

Two findings generalise beyond this application. First, a calibration advantage that appears substantial from a single training run (a 40% ECE reduction) is roughly halved and becomes seed-fragile under five-seed replication; single-seed calibration claims should be treated with caution, and multi-seed replication should be standard. Second, where clinical inputs are heavily and non-randomly missing, a multimodal model can exploit missingness structure and present a spurious benefit that neither reflects the measured values (the lab-present subcohort shows no value effect) nor constitutes leakage (temporal, split and preprocessing audits are clean). The missingness-only ablation makes this mechanism explicit and is, to our knowledge, a rarely reported control in clinical multimodal studies. Both cautions are actionable and inexpensive.

When should fusion be expected to help? Our result sharpens a practical heuristic. Multimodal CXR fusion has been most successful for *prognostic* and *screening* targets whose ground truth is not the radiograph itself, for example ICU mortality (Hayat et al, 2022), inpatient outcomes (Soenksen et al, 2022) and ED acute-heart-failure screening (Lee et al, 2024), where clinical context supplies information the image cannot. Radiographic pneumonia is the opposite case: the label is defined by the image, so the marginal information in triage vitals or early labs is small, and our well-powered negative result is exactly what this heuristic predicts. The finding is thus not that fusion "fails" but that its value is conditional on the target lying partly outside the imaging modality, which should guide where fusion is worth the added complexity in ED decision support.

Comparison to related work. Prior multimodal CXR studies typically report single-run gains on ICU or inpatient cohorts (Hayat et al, 2022; Soenksen et al, 2022; Lee et al, 2024); we instead evaluate ED pneumonia under a verified time-safety constraint with multi-seed replication and explicit audits, and reach a negative but well-powered conclusion on discrimination. Our missingness finding complements the informative-missingness literature: where GRU-D and related models *use* missingness as a predictive feature (Che et al, 2018), we show that in a fusion setting the same signal can be mistaken for a chemistry-driven gain, and we provide a simple flags-only control that disentangles the two. Our calibration finding aligns with benchmarks showing that single-model uncertainty is unreliable under distribution shift while model averaging is more robust (Ovadia et al, 2019). Direct numerical comparison across cohorts and label definitions is uninformative; the contribution is the rigour of the evaluation and the two transferable cautions.

Limitations. Training and multimodal evaluation are single-institution (MIMIC) and retrospective; external validation confirms the image branch generalises to NIH but is necessarily limited to imaging, as no external dataset links ED triage to the CXR, and probabilities require recalibration to the target base rate before deployment (the *u_ignore* policy also raises internal evaluation prevalence to 45.3%). The lab-present subcohort is small (n=269); the modern-backbone comparison uses frozen features rather than end-to-end fine-tuning; and NIH labels are NLP-derived, adding label noise to the external estimate.

6 Conclusion

Under strict time-safety and multi-seed evaluation, fusing immediately available clinical context with the chest radiograph does not reliably improve pneumonia discrimination, and the one apparent gain is an artefact of missing-data structure. The image model itself is robust and generalises across institutions with an AUROC drop of only 0.016. Beyond the negative result, the study contributes a reproducible, time-safe multimodal ED pipeline and two reusable cautions (single-seed calibration inflation and missingness-as-signal) for the growing field of clinical multimodal AI. Future work includes end-to-end fine-tuning of modern foundation backbones, multi-site validation, and prospective evaluation with recalibration to the local base rate.

Declarations

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Authors' contributions. YAD designed and implemented the data pipeline, trained all models and performed the analyses. AME conceptualised and supervised the study and contributed to the manuscript. MM and SS contributed to methodology, interpretation and critical revision. All authors read and approved the final manuscript.

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Ethics approval. All data are publicly available and de-identified (MIMIC under PhysioNet's credentialed Data Use Agreement; NIH ChestX-ray14 public release). No additional ethical approval was required; procedures followed the Declaration of Helsinki.

Data and code availability. MIMIC and NIH ChestX-ray14 are publicly available (MIMIC via PhysioNet credentialing). All pipeline, training and evaluation code is available at `https://github.com/wa5rx3/MultiModalPneumonia`; every reported number is regenerated by a named script tied to the fixed temporal split.

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